NHS Scotland A&E Performance Dashboard: Requirements

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Purpose

The dashboard shows where and why A&E departments in Scotland are missing the 4-hour standard, so performance teams can target action at the hospitals with the biggest gaps.

The standard: 95% of patients attending A&E should be admitted, transferred or discharged within 4 hours of arrival. Waits of more than 8 and 12 hours are also tracked, as they carry the highest risk to patients.

The need: performance data is published monthly by Public Health Scotland, but as large tables. Managers have no single view that compares health boards and hospitals, separates major emergency departments from minor injury units, and shows whether long waits are getting better or worse.

Scenario: this is a portfolio project built on real, published data. The stakeholder roles below are typical of an NHS unscheduled care performance team; no real individuals are represented.

Stakeholders and users

Three groups use the dashboard, each at a different level of detail, and each health board sees only its own hospitals.

User	What they need	Data they can see
National performance lead (sponsor)	Scotland-wide performance, comparison across all 14 health boards, trends over time	All health boards
Health board director of unscheduled care	Their board against the standard and against other boards, which of their hospitals are struggling	Their own board in detail, plus the Scotland total for comparison

User	What they need	Data they can see
Hospital site manager	Their site's monthly performance, long waits, and how it has changed	Their own board's hospitals

Business questions

The dashboard must answer eight questions; every page and measure traces back to at least one of them.

1. **How is Scotland performing against the 95% standard,** this month and over time?
2. **Which health boards and hospitals are furthest below the standard,** and by how much?
3. **Are waits of more than 8 and 12 hours rising or falling,** and where are they concentrated?
4. **How do major emergency departments (Type 1) compare with minor injury units (Type 3)?**
5. **Is performance linked to demand?** Do sites perform worse in months with more attendances?
6. **Is there a seasonal pattern,** for example a winter dip, and how deep is it?
7. **Which sites have improved or declined most** over the last 12 months?
8. **How complete is the data?** Which sites lack the patient-level data needed for 8 and 12-hour figures?

Data sources and scope

All data is published by Public Health Scotland under the UK Open Government Licence and covers every A&E site in Scotland, monthly, from July 2007.

Source	What it provides	Link
Monthly A&E Activity	Attendances and waits within 4 hours, over 8 hours and over 12 hours, by month, health board, hospital, department type and attendance category	Monthly A&E Activity and Waiting Times
Current NHS Hospitals in Scotland	Hospital names and the health board each belongs to	Hospital Codes

Source	What it provides	Link
Health Board 2014 to 2019	Health board names for each board code	Public Health Scotland open data portal

In scope

- All A&E sites, both major emergency departments (Type 1) and minor injury units (Type 3)
- Full history from July 2007, with dashboard pages defaulting to the last five years
- Monthly totals only

Out of scope

- Patient-level records, reasons for attendance and diagnoses
- Weekly data, demographics, referral source and discharge destination (possible later additions)
- Staffing, bed occupancy or other causes outside the A&E data

Two data rules that prevent wrong totals

- **Attendance category:** each site and month appears as "Unplanned", "New Planned" and "All" (the two combined). Totals use **"All" only**, which matches the revised 4-hour standard that has included planned attendances since February 2025.
- **Department type:** Type 1 and Type 3 are reported separately by default, as minor injury units almost always meet the standard and can hide problems in emergency departments.

KPI definitions

Every percentage is recalculated from counts at the level being viewed, never averaged, so a small minor injury unit can't count as much as a large emergency department.

KPI	Definition	Calculation	Answers question
Attendances	Number of A&E attendances	Sum of NumberOfAttendancesAll	1, 5
% seen within 4 hours	Share of attendances completed within 4 hours	Sum of NumberWithin4HoursAll ÷ sum of NumberOfAttendancesAll	1, 2, 4

KPI	Definition	Calculation	Answers question
Gap to standard	How far performance sits from 95%	% seen within 4 hours minus 95%, in percentage points	2
Patients waiting over 4 hours	Number of attendances not completed within 4 hours	Sum of NumberOver4HoursAll	2
% waiting over 8 hours	Share of attendances lasting more than 8 hours	Sum of NumberOver8HoursEpisode ÷ sum of NumberOfAttendancesEpisode	3
% waiting over 12 hours	Share of attendances lasting more than 12 hours	Sum of NumberOver12HoursEpisode ÷ sum of NumberOfAttendancesEpisode	3
Change over 12 months	Movement in % seen within 4 hours	This month's % minus the same month last year, in percentage points	7
Months meeting standard	How often a site meets 95%	Count of months where % seen within 4 hours is 95% or more	2, 7
Episode data coverage	Share of attendances with patient-level data	Sum of NumberOfAttendancesEpisode ÷ sum of NumberOfAttendancesAll	8

Calculation rules

- **Never average percentages.** Always divide summed counts, at whatever level the user is viewing.
- **8 and 12-hour measures use the episode columns.** Only sites that submit patient-level data report these, so they're divided by episode attendances, not all attendances.
- **Quality flags:** a "z" flag means the figure isn't available for that site. These are stored as empty values, not zeros, so they don't pull percentages down.
- **Default filter:** attendance category "All", department type shown separately.

Dashboard pages and requirements

Five pages, each answering a set of the business questions, with the summary page first.

Page	What it shows	Answers question
Executive summary	Latest month's % within 4 hours against 95%, attendances, long waits, and the five-year trend	1, 3
Health board comparison	All 14 boards ranked by gap to standard, with 12-month change	2, 7
Hospital detail (drill-through)	One hospital's monthly trend, long waits and months meeting the standard	2, 3, 7
Demand and seasonality	Attendances against performance, and the average pattern by month of year	5, 6
Data quality	Episode data coverage by site, and sites missing 8 and 12-hour figures	8

Functional requirements

- [ ] Filters for month, health board, hospital and department type on every page
- [ ] Right-click drill-through from any hospital to its detail page
- [ ] Row-level security: a health board user sees only their own board, with the Scotland total for comparison
- [ ] The 95% standard shown as a reference line on every trend chart
- [ ] Colours that show at a glance whether a figure meets the standard
- [ ] A "data last updated" date and the Public Health Scotland source on every page

Data quality, refresh and acceptance

The dashboard is accepted once its Scotland figures match Public Health Scotland's published numbers for sample months.

Refresh

- Public Health Scotland publishes monthly, about two months in arrears.
- Refresh after each monthly release: replace the source file, rerun the SQL load, refresh Power BI.
- Revisions to past months are applied on each refresh, as the full history is reloaded.

Data quality checks (run in SQL on every load)

- [ ] No duplicate rows for the same month, hospital, department type and category
- [ ] Within 4 hours plus over 4 hours equals total attendances
- [ ] Every hospital code matches a hospital name, and every board code matches a board name
- [ ] No month missing from the date range
- [ ] "z" flags converted to empty values, not zeros

Acceptance criteria

- [ ] Scotland % within 4 hours matches the published figure for three sample months
- [ ] A health board user sees only their own board's hospitals
- [ ] Changing any filter updates every visual on the page
- [ ] Each of the eight business questions can be answered from the dashboard